A Gaggle of Transformers Under a Graded-Where Harness

Continual learning by consolidating interventions, with the parameter count held down by the causal structure the harness already owns

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Abstract

The inhabiting-choice note established that a transformer trained on completed runs is a rung-one object: a trace distribution has a unique autoregressive factorisation, so branching structure is invisible to any learner that does not act, and the agent loop around the model is where the acting happens. The interventions-at-the-cut note then supplied the acting: a scoped `do` on the cut of the rho calculus with graded where-clauses, under which Pearl's calculus embeds exactly and a refusal is a receipt that starves. This note puts the two together as an engineering proposal. A *harness* written in rho with graded where-clauses and `do` performs experiments on a population of specimens; the experiments produce traces that carry their intervention, their exogenous record, and their starvations; and the traces are consolidated into a *gaggle* of small transformers, each of which is itself a process whose attention is a graded race in the log semiring. The claim defended is that a gaggle *indexed by interventions* is a rung-two object even though every member of it is rung-one, because the family of interventional distributions over a staged fragment determines the causal model, and that the harness, being a rho term, is a witness in which the hedge is inherited — which is what moves the design off the saddle point the inhabiting-choice note located, to the extent it moves at all. The parameter-count question gets a structural answer rather than a compression trick: every gadget of the embedded causal model needs a mechanism, not a world model, so a transformer in the gaggle learns $P_i(v \mid pa)$ over a parent set that is syntactic, with a context window the size of that parent set and no obligation to represent the exogenous noise the resolver already owns. Four further levers follow from the companion notes: factorisation of the gaggle along namespaces the do-calculus proves d-separated; pruning of attention heads by intervention on the transformer's own races; consolidation of members that are *interventionally* bisimilar, a test the harness can afford because it owns their code; and compilation of a member out of the gaggle altogether into an $\mathcal{F}_1$ clause when its learnt mechanism is a bounded-depth posynomial, at which point the accumulated science lives in the clause language and the transformer was scaffolding. The note closes with the algorithm as a rho sketch, the execution story on f1r3node and Saph1r3, and seven obligations, of which the first — that the intervention-indexed family is rung two — is the one the rest depend on.

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1 The question

The three companion notes draw a line, and the present note is an attempt to build on the right side of it.

The first [1] observes that a system whose witnesses reduce confluently can state branching phenomena but cannot inhabit them, and reads the observation for artificial intelligence: a distribution over finite traces has a unique autoregressive factorisation, so two systems that differ only in *when* a choice was made — Milner's $\mathit{coin}.(\mathit{tea} + \mathit{coffee})$ against $\mathit{coin}.\mathit{tea} + \mathit{coin}.\mathit{coffee}$ — induce the same trace distribution and are indistinguishable to any learner whose only access is samples. The observable that separates them is a refusal, and refusals do not appear in logs of completed runs. A next-token predictor is therefore rung one of Pearl's ladder by an information-theoretic limitation; the agent loop around it supplies rung two from outside; and the arrangement — confluent predictor, branching in the harness — is a saddle point, stable as long as per-query answers are what is paid for and unstable the moment what is wanted is an accumulating, heritable body of search structure.

The second [2] supplies the construct the harness needs in order to intervene at all: $\mathsf{do}(C)\ \mathsf{in}\ P$, a scoped constraint on the bindings of a race, placed at the cut because whether a cut will occur is undecidable, lowered into the graded where-clause by the normal-form theorem for weight maps, and exact on the staged fragment as Pearl's graph surgery. Under it the vending-machine refusal is a receipt that starves, counterfactuals are replay from a recorded pre-state, and the three rules of the do-calculus are theorems about the embedded fragment.

The third family [6, 7, 8, 9] is the route from a rho term to a data-parallel device, through name-free combinators to an incidence matrix of atoms against names, on which redex finding is a join, conflict detection a second sparse product, and a machine step a maximal independent

set of the conflict graph. The route is strictly classical today — no weights — and the conflict components it already computes are the contention sets at which the graded semantics normalises.

The question this note asks is the one a working engineer asks after reading the three: *granted that a transformer is rung one and a harness in rho with* do *is rung two, what is the right way to lash several transformers into such a harness so that the harness's experiments make the transformers better, and so that the transformers stay small?* The emphasis on size is not incidental. A transformer that must model the whole world is large because the world is; a transformer that must model one mechanism of a causal model the harness has already factored is small because the mechanism is. The size question and the rung question turn out to have the same answer.

1.1 The shape of the proposal

- A *harness* $\mathcal{H}$ is a rho term with graded where-clauses and do, confined to a namespace it owns ($\mathcal{T}_4$ of [2]), holding a budget. It performs *experiments*: it chooses a specimen, a pre-state, and a $\mathsf{do}(C)$, runs the intervened term, and records what happened.
- A *trace* is not a log of completed actions. It is a tuple $(S, C, h, u, \star)$: the pre-state, the intervention, the causal history, the exogenous record (which candidate fired at each site), and the starvation flag. A trace carries its own intervention, which is what makes it rung-two data rather than rung-one data (§3).
- A *gaggle* $\mathcal{G} = \{T_1, \ldots, T_k\}$ is a population of small transformers, each a process in the sense of the earlier transformer proposal [5]: attention is a graded receipt in the log semiring, weights rest on channels, and the whole is a term the harness can quote, intervene upon, and fund.
- *Consolidation* is the loop that takes traces to transformers: routing a trace to the member that should learn from it, retraining that member or spawning a new one, and then *shrinking* the gaggle by merging members the harness can prove interventionally equivalent and by compiling members out into clauses when their learnt mechanism admits one.

1.2 What the note claims, and what it concedes at once

The claim is that this arrangement is a rung-two learner whose knowledge is inherited, and that its parameter count is governed by the causal structure the harness learns rather than by the size of the world.

The concession is that every transformer in the gaggle is, on its own, exactly the rung-one object the inhabiting-choice note describes, and that the branching lives in the harness. The note's answer to the saddle-point charge is not that the charge is wrong; it is that the harness is a rho term, hence a witness that can be a race, hence inheritable, and that the design's job is to move as much of the accumulated science as possible from the transformers (confluent, not inheritable as races) into the harness's clauses (races, inheritable). §7 says exactly how far that goes.

2 What the context fixes

The note is self-contained in its definitions but not in its theorems; the following are imported.

Definition 1 (Graded receipt, contention, likelihood [3])**.** A receipt $\mathsf{for}(\vec{y} \leftarrow \vec{x}\ \mathsf{where}\ \psi)P$ is a join with a clause ψ valued in a commutative semiring $\mathbb{V}$, evaluated once per candidate match. Two candidates contend when they share a message or a linear receipt; a contention set is a

connected component; resolution normalises within it. Over $\mathbb{R}_{\geq 0}$ the likelihood of a causal history $h = (\Gamma_i, \sigma_i)_i$ is $\prod_i w(\sigma_i) / \sum_{\tau \in \Gamma_i} w(\tau)$, and the parameter gradient of its logarithm is a sum of local scores, one per firing site.

Definition 2 (The do membrane [2])**.** $\mathsf{do}(y_1 \models \varphi_1 \,\&\, \cdots \,\&\, y_n \models \varphi_n)$ in P reads every receipt in scope binding y_i with clause $\psi_r \otimes [\![\varphi_i]\!]$, normalising after the restriction. On the flat fragment it lowers exactly into the where-clause. On the staged fragment $\mathcal{T}_1$ every receipt is its own contention set, the causal order is a static DAG, and do on a gadget's ancilla binding is Pearl's graph surgery; positivity is the absence of starvation; counterfactuals are replay with the exogenous record held.

Proposition 3 (Observational invisibility [1, Prop. 4.1])**.** *Two labelled transition systems inducing the same trace distribution induce the same autoregressive factorisation; a learner with access only to samples cannot distinguish them.*

Proposition 4 (Interventions have one carrier [2, §3])**.** *A site with a single candidate is intervention-transparent; the observation/intervention gap lives only on races; over $\mathbb{R}_{\geq 0}$ a context cannot force a race it does not own.*

Observation 5 (The three requirements [1, Obs. 5.1])**.** *An improvement operator capable of recursion must (i) be defined on witnesses that are races, (ii) preserve branching through evaluation, and (iii) fund branches that lose at a rate that is itself subject to the operator.*

Definition 6 (A transformer as a process [5])**.** A decoder-only transformer with L layers and H heads is the rho term in which the residual stream at (ℓ, i) rests on a channel, keys and values at (ℓ, h) rest as messages read by peek, and each query is a linear graded receipt

$$\mathsf{for}(\, @j, @k, @v \mathrel{\triangleleft\!\!-} @[\mathrm{kv}, \ell, h] \text{ where } [\![j \leq i]\!] \otimes \tfrac{1}{\sqrt{d}}\, q \cdot k \,)\ P(v),$$

read in the log semiring $(\mathbb{R} \cup \{-\infty\}, \mathrm{logaddexp}, +)$, so that normalisation within the query's contention set is softmax and the tropical limit is argmax. Under the path-sum reading, collapsed at the affine boundary before the next nonlinearity, the receipt computes soft attention; under the stochastic reading it computes stochastic hard attention with the same expectation. The dense arithmetic is in the ground layer; the only race in the term is attention.

Definition 6 is what makes the rest of the note possible: the transformer is a term, so the harness can quote it, run it under a do, and read its exogenous record, and by Proposition 4 the only sites at which an intervention on it has content are its attention heads.

3 The data: traces that carry their intervention

The inhabiting-choice note remarks that "logs that record failed attempts make the refusal a trace symbol and the distinction learnable, but such logs are intervention data in disguise, which is the point rather than a loophole." The present design takes the remark as its data schema.

Definition 7 (Experimental trace)**.** An *experimental trace* is a tuple $t = (S, C, h, u, \star)$ where S is a pre-state (a closed term, or its hash in the shard's state store), C is a constraint in $\mathcal{F}_0$ or $\mathcal{F}_1$, h is the causal history of $\mathsf{do}(C)$ in S to its normal form or budget, u is the exogenous record (the fired class at each site), and $\star \in \{\mathrm{ok}, \mathrm{starved}\}$ records whether some restricted receipt starved. The *observational projection* of t is the sequence of readouts along h; it is what a log of completed runs would contain.

The intervention C is part of the datum, and so is the starvation flag. A transformer trained on experimental traces is trained to predict h *given* (S, C), and that is a different object from a transformer trained on observational projections.

Proposition 8 (An intervention-indexed family is rung two)**.** *Let M be a positive causal Bayesian network over finite domains embedded as the staged term $[\![M]\!]$, and let $D(C)$ denote the distribution of observational projections of $\mathsf{do}(C)$ in $[\![M]\!]$. Then the family $\{D(C) : C \in \mathcal{F}_0\}$ determines M up to the equivalence of causal Bayesian networks, whereas $D(\top)$ alone determines M only up to Markov equivalence. In particular the family separates the two vending machines of [1, §4.6], which $D(\top)$ does not.*

Proof sketch. By [2, Thm. 5.3], $D(C)$ for C a point restriction on gadget i is the distribution of the submodel $M_{V_i=v}$. The joint distributions of all single-variable hard interventions determine the DAG up to the standard result that the interventional family identifies the graph when every variable can be intervened on, and the conditional tables are read off the intervened distributions directly. For the vending machines, $D(y = \mathit{tea})$ has a starved trace with positive probability for Q and none for P, and the starvation flag is part of the datum. □

The proposition is stated on the staged fragment because that is where the companion note's theorems hold; Obligation 1 asks for the general statement. What it already says is enough to fix the design: *the harness must choose C, and must choose it well.* The quality of the gaggle is bounded by the coverage of the interventions in its training set, and the harness's job is experimental design, not data collection.

Remark 9 (Which rung the transformer is on)*.* Each member of the gaggle remains a conditional trace predictor and is rung one in the precise sense of Proposition 3: given (S, C) it predicts a trace distribution, and that distribution has a unique factorisation. What is rung two is the *family*, indexed by C, and the index is supplied by the harness. This is the saddle point, named exactly, and §7 is where it is addressed.

4 Where a transformer sits in a causal model

The embedding of [2] places, at each variable V_i, a gadget whose clause is the conditional probability table $P_i(v \mid pa)$ in disjunctive normal form. On finite domains of modest size that clause is a lookup. A transformer earns its place only where the lookup is unaffordable: large or structured domains, mechanisms that are compositional in their inputs, or variables whose values are themselves terms.

Definition 10 (Transformer-backed gadget)**.** For a variable V_i with parents $V_{p_1}, \ldots, V_{p_m}$, a *transformer-backed gadget* replaces the enumerated clause by

$$\mathsf{for}(\, p_1 \leftarrow v_{p_1}^{(i)} \,\&\, \cdots \,\&\, p_m \leftarrow v_{p_m}^{(i)} \,\&\, u \leftarrow w_i \;\mathsf{where}\; [\![\mathrm{dom}(*u)]\!] \otimes \mathsf{T}_i(*p_1, \ldots, *p_m)[*u]\,)\ \mathit{emit}_i(*u),$$

where T_i is a member of the gaggle run as a pure ground expression whose output is a vector of log-weights over D_i, indexed by the ancilla value. The ancilla channel w_i carries D_i or, for large domains, is sampled from the gaggle member's own vocabulary race.

Three things about this placement decide the parameter question.

Observation 11 (The context window is the parent set)**.** *The clause at gadget i reads pa_i and nothing else, because the parent set is syntactic [2, §1.2]: the names the pattern and the clause read. A transformer backing gadget i therefore has input length $|\mathrm{PA}_i|$ (in tokens, the serialisation of the parent values), not the length of the history. Attention's job of discovering which earlier positions matter has been done by the harness's experiments and is recorded in the term as the pattern; what remains for the transformer is the mechanism.*

Observation 12 (The transformer does not model the noise)**.** *The exogenous noise U_i is the resolver's draw at the ancilla race, and it is the harness's, not the transformer's. A transformer trained on observational projections must represent the correlations the noise induces across variables — confounding — because it cannot tell them from mechanism. A transformer backing a gadget predicts $P_i(v \mid pa)$ and is* forbidden *by its placement from seeing anything else. The parameters that an observational model spends on confounders are not spent.*

Observation 13 (Only the races are learnable by intervention)**.** *By Proposition 4, an intervention at a deterministic site is either the identity or disables the site. The harness's experiments therefore inform exactly the gadgets that are races — the ancilla receipts — and the deterministic plumbing between them (fan-out, readout, the transformer's own projections and MLPs) is intervention-transparent. This is not a limitation on what the gaggle learns; it is a statement about where the gradient lands. The score of [3, Thm. 4.3] is a sum over firing sites, and a deterministic site contributes zero.*

Remark 14 (Activation patching is a do). The interpretability community's causal tracing and activation patching — hold a run, replace one head's output with that from a corrupted run, measure the change downstream — is, in the vocabulary of this programme, replay under $\mathsf{do}(v \models \cdot)$ at an attention site with the exogenous record held, which is [2, Thm. 5.6] applied to the transformer's own term. That the technique works on heads and is ill-defined on MLPs is Proposition 4: heads are races, MLPs are not. The harness therefore has, for free, an instrument for interrogating its own gaggle members that the community built by hand, and §6 uses it to prune.

5 The consolidation loop

5.1 Overview

The loop has five stages, each of which is a graded race in the harness term and so is itself subject to the harness's budget and to intervention.

1. **Propose.** Choose a specimen S and a constraint C. The choice is a graded race whose clause is the harness's current estimate of the information the experiment would yield; §5.2.
2. **Run.** Execute $\mathsf{do}(C)$ in S under the budget; record the experimental trace t of Definition 7. On f1r3node this is one deployment whose trace the validators already produce.
3. **Route.** Deliver t to the gaggle member that should learn from it. Routing is a graded receipt whose candidates are the members and whose clause is each member's affinity for t; §5.3. If no member accepts, the receipt starves, and starvation is the signal to spawn.
4. **Revise.** Retrain the chosen member on its accumulated traces, with the PLN natural-gradient discipline of [3] so that confidence is the metric; §5.4.
5. **Shrink.** Merge members the harness can prove interventionally bisimilar; prune heads by intervention; compile a member out into a clause when its mechanism admits one; let members that no longer earn their keep starve; §6.

5.2 Propose: experimental design as a graded race

The harness holds, for each gadget i of its current model, a graded clause ψ_i (enumerated or transformer-backed) and an evidence count n_i in the PLN sense. A candidate experiment is a pair (S, C) with C a point or set restriction on some gadget's ancilla. Its value is placed in the clause of the proposal receipt:

```
for( @S <<- specimens & @C <<- candidates
     where  [ C in F0 and reachable(S, site(C)) ]
            * gain(S, C)                        // expected score, Eq. (1)
            * 1 / (1 + cost(S, C))              // phlo to run it
   ) { run!(S, C) }
```

The gain is the expected magnitude of the local score the trace would contribute at the intervened site — the quantity [3, Cor. 5.7] accumulates, available before the experiment is run because the candidate weights the site would evaluate are the harness's own clause. Two remarks. First, the proposal receipt is itself a race, so the harness hedges its experiments: under the stochastic reading it does not always run the single most informative one, and the fraction of budget spent on lower-gain experiments is a clause constant the harness can tune — requirement (iii) of Observation 5 applied to experimental design. Second, the reachability guard is $\mathcal{F}_2$ in general and the vacuity question of [2, §7.1]; under $\mathcal{T}_1$–$\mathcal{T}_3$ it is decidable by inspection and is the positivity check.

5.3 Route: the gaggle as a mixture of experts whose gate is a clause

Members rest on a channel as quoted terms together with a *signature* sig_j: a crisp formula over traces describing which traces the member has been trained on — in the simplest case, the set of sites its gadgets back. Routing is

```
for( @t <<- traces & @(j, sig, T) <<- gaggle
     where  [ sig(t) ] * affinity_j(t)
   ) { revise!(j, t) }
```

where $\mathrm{affinity}_j(t)$ is the likelihood member j assigns to t's history under t's intervention, evaluated by running T_j as a pure expression. Read in the log semiring this is a softmax gate over experts, which is to say that the gaggle's routing *is* attention over members, with the same clause that attention within a member uses. Three consequences:

- A trace no member's signature admits leaves the routing receipt with no candidate of nonzero weight. The receipt starves. Starvation here is not death but birth: a second receipt waits on the same channel with clause $[\![\neg \bigvee_j \mathrm{sig}_j(t)]\!]$ and spawns a fresh member with signature generated from t's site. The gaggle grows exactly when the world presents a mechanism it has no expert for, and the *refusal* of the gaggle to absorb a trace is the growth signal — positivity as starvation, read constructively.
- A trace several members admit is a race between them, and under the stochastic reading more than one may be trained on it over time, which is the hedge: the harness funds the losing expert at a rate set by the clause.
- The signatures are crisp formulae in the term and are therefore visible to the model checker, so "is there a mechanism two members both claim?" is a spatial question the harness can ask of itself, and it is the trigger for the merge test of §6.4.

5.4 Revise: retraining without losing plasticity

A member retrained on its bucket faces the classical continual-learning hazard: new traces overwrite old mechanisms. The companion paper measured the same hazard in a different guise. Two learners circulating belief to one another inflate their evidence counts without adding information, and the pathology is *loss of plasticity rather than instability* [3, §11]. The natural-gradient result there — revision is a Fisher-metric step with the metric evaluated at

the posterior count, hence implicit and unconditionally stable — is the discipline the gaggle should adopt, and it fixes two design rules.

Principle 15 (Conservation of evidence in the gaggle)**.** A member is never trained on a trace it generated. Traces are produced by experiments on *specimens*; a member's own outputs, when the harness runs it as a specimen (§6.4), are tagged as testimony and contribute to no member's evidence count. Without this rule the gaggle inflates its own confidence by re-reading itself, and the measured consequence is a gaggle that can no longer learn.

Principle 16 (Confidence is the step size)**.** Each member carries, per gadget it backs, the PLN count n_i, and its learning rate on traces at site i is the Robbins–Monro schedule that count induces. A member backing a well-established mechanism takes small steps; a fresh member takes large ones. The count is data on a channel and the harness may intervene on it — resetting a count is the do that restores plasticity.

Both principles are the companion paper's results read for a population of transformers rather than for a single clause, and neither requires anything a transformer cannot do: the Fisher metric of a softmax output is standard, and the per-site counts are bookkeeping.

6 Keeping the parameters down

The brief asked specifically for the ability to keep the transformers small. The answer has one structural part and four operational parts, and all five are consequences of the companion notes rather than additions to them.

6.1 The structural part: mechanisms, not worlds

Proposition 17 (Parameter bound on the staged fragment)**.** *Let M be a causal Bayesian network over n variables with domains $D_1, \ldots, D_n$ and parent sets $\mathrm{PA}_1, \ldots, \mathrm{PA}_n$. A gaggle backing M on the staged fragment needs, in total, parameters sufficient to represent*

$$\sum_{i=1}^{n} \left| \prod_{j \in \mathrm{PA}_i} D_j \right| \cdot |D_i|$$

table entries — the size of the conditional probability tables — and a transformer is required at gadget i only when that product is unaffordable as a lookup. An observational model of the same system must represent the joint, of size $\prod_i |D_i|$, less whatever factorisation it can infer from observational data alone, which is the Markov-equivalence class and not M.

Proof. The first statement is Theorem 5.2 of [2]: the likelihood is the truncated factorisation, and each factor is one gadget's clause. The second is the standard gap between a joint and its CBN factorisation, with Proposition 8 supplying that interventions recover the factorisation and observation does not. □

The proposition is almost a tautology, and that is its value: the parameter count of the gaggle is the parameter count of the *model*, in the sense in which Pearl uses the word, and the harness's experiments are what establish the model. The gaggle is small when the world is modular, and the harness finds out whether it is.

6.2 Lever 1: factorise along namespaces

The composing-learners note [10] established that a learner is a namespace and that disjointness of namespaces is factorisation of the distribution. The do-calculus gives the harness a

test for disjointness it can run: if $(Y \perp\!\!\!\perp Z \mid X)$ holds in the appropriate mutilated graph, rule 1 of [2, Cor. 5.4] says the mechanisms on Y and Z may be learnt separately, and the harness should keep them in separate members. The interventions note's first open problem — d-separation as a crisp behavioural formula over the staged term [2, §10] — is therefore load-bearing here: when it is discharged, "may these two members be kept apart?" is a question the model checker answers, and the gaggle's factorisation is maintained as a checked invariant rather than a hope.

Two members on d-separated namespaces have, between them, the parameters of two mechanisms. One member on their union has the parameters of a mechanism over the product. The lever is to never pay for the product when the harness has proved it need not.

6.3 Lever 2: prune heads by intervention

A member's attention heads are its only races (Definition 6), so they are the only sites at which the harness's do has content on the member itself. For each head (ℓ, h) of member T_j the harness runs

$$\mathsf{do}(v \models [\![v = 0]\!]) \text{ in } T_j \quad \text{at every query of head } (\ell, h)$$

— restrict the head's value race to the zero vector — over a held-out set of experimental traces, and compares the member's interventional likelihoods before and after. A head whose removal leaves the likelihoods within tolerance on every intervention in the held-out set is pruned. This is ordinary head pruning; what the setting adds is that the comparison is over *interventional* distributions, so a head that matters only under interventions the observational data never exhibited is kept, and one that matters only for modelling confounders the member need not model is dropped. The tropical limit of the same clause — argmax attention — is the quantisation step of [5] and composes with pruning.

6.4 Lever 3: merge members that are interventionally bisimilar

The inhabiting-choice note's thesis, applied to the gaggle's own members, says that trace equivalence between two members is the wrong test for merging them: two members may assign identical likelihoods to every observational projection and differ on what they do under an intervention. The right test is bisimulation, and the harness can afford it because it owns the members' code ($\mathcal{T}_4$) and can run the bisimulation game on them.

Definition 18 (Interventional bisimilarity of members)**.** Members T_j and T_k with the same signature are *interventionally bisimilar to tolerance ϵ on a constraint set $\mathcal{C}$* when for every $C \in \mathcal{C}$ and every pre-state S in the held-out set, the distributions of experimental traces of $\mathsf{do}(C)$ in $T_j(S)$ and $\mathsf{do}(C)$ in $T_k(S)$ are within ϵ in total variation, starvation flags included.

The test is the Larsen–Skou probabilistic bisimulation game [17] with the harness as the tester, restricted to a finite constraint set because the general question is undecidable, and the restriction is honest: the members are bisimilar *as far as the harness has been able to experiment*, and the merge is an induction from that evidence with the evidence count attached. Merging means retaining one member and deleting the other, or distilling both into a fresh member of smaller size trained on the union of their buckets under Principle 15 (the distillation targets are testimony, counted at zero).

The reason to prefer this test to the obvious one — compare the members' weights, or their outputs on a validation set — is Proposition 3 once more. Output agreement is trace equivalence. Two members that agree on every completed run and differ on a refusal are different mechanisms, and merging them discards one.

6.5 Lever 4: compile a member out into a clause

The strongest lever is the one that removes a transformer entirely.

Proposition 19 (Compile-out)**.** *Let T_j back gadget i and suppose its learnt mechanism $pa \mapsto P_i(\cdot \mid pa)$ is representable, to tolerance ϵ on the interventional held-out set, by a graded formula $\psi \in \mathcal{F}_1$ of depth d over the parent data. Then replacing the transformer-backed gadget by the enumerated gadget with clause ψ preserves every interventional distribution of the model to tolerance ϵ, costs $O(2^d)$ per candidate at the site rather than a forward pass, and reduces the member's parameter count to the constants of ψ.*

Proof. The gadget's contention set is itself ($\mathcal{T}_1$), so its contribution to every likelihood is its normalised clause value at the fired class, and two clauses within ϵ on every parent value contribute within ϵ to every factor. The cost figure is [2, Def. 7.2]. □

How to find ψ is the research question, and the companion notes already own half of it. An $\mathcal{F}_1$ graded formula over $\mathbb{R}_{\geq 0}$ is a posynomial in its constants [3, Prop. 5.2]; fitting a posynomial of bounded depth to a transformer's conditional outputs is a symbolic-regression problem over a grammar the omnibus [15] generates from the term language, and the crisp subformulae are spatial tests on the parent data that the member's attention pattern suggests. The proposal is to run compile-out as a standing race in the harness: for each transformer-backed gadget, a candidate ψ is proposed from the member's attention structure, tested interventionally as in §6.4, and adopted if it passes, with the budget for the attempt a clause constant.

What compile-out buys is more than parameters. A clause is a term; it is quotable, inheritable by the reproduction note's crossover [12], checkable by the model checker, and *a race*. A transformer's weights are none of these. Every mechanism the gaggle manages to compile out has moved from the confluent side of the line to the side where the accumulated science lives as witnesses, which is the inhabiting-choice note's whole demand. The transformers are scaffolding for mechanisms the harness has not yet understood well enough to write down.

6.6 Lever 5: mortality

A member is funded from the harness's budget at a rate proportional to the score its traces contribute. A member that backs a gadget no experiment has touched within a window, or whose traces the merge test has shown another member to cover, starves. This is the deadlock-by-starvation pruning of [13] applied to the gaggle and needs no coordinator; the harness's budget is conserved, and a member that does not earn tokens cannot pay for its retraining. The funding rate for losing members — those that lose routing races — is a clause constant and is itself in the harness's parameter set, which is requirement (iii) of Observation 5.

7 The saddle point, revisited

The inhabiting-choice note's charge against the loop architecture is precise: the hedge lives in the harness, is collapsed at the boundary, and is handed to no one as a race. The design above is a loop architecture and must answer the charge on its own terms.

What is still on the saddle. Each member is a confluent predictor. Its parameters are improved by a gradient on outcomes, and the diversity its next improvement would need is what the current improvement consumes, exactly as [1, §5.4] says. Nothing here changes that about the transformers.

lever	what it removes	what licenses it
mechanisms, not worlds	parameters spent on the joint and on confounders	truncated factorisation; the parent set is syntactic
factorise	the product of two mechanisms when the sum suffices	do-calculus rule 1; d-separation as a checked formula
prune heads	heads that matter under no intervention	interventions have one carrier; heads are the races
merge	duplicate members	interventional bisimilarity, a test the harness can run
compile out	the transformer itself	$\mathcal{F}_1$ clauses are posynomials; exact on $\mathcal{T}_1$
mortality	members that earn nothing	conserved budget; starvation as pruning

Table 1: The parameter levers.

What has moved. Three things.

1. The harness is a rho term. Its proposal race, its routing race, its compile-out race are witnesses that are races, and its funding rates for the losing branches of each are clause constants in the term. The harness therefore meets the three requirements of Observation 5 *for itself*: it can be quoted, inherited with its hedges uncollapsed, and improved by an operator that acts on its clauses. The desert annual's germination fraction is here the routing clause's constant for losing experts.
2. The gaggle is not a set of outcomes but a population of terms resting on a channel, and the term that holds it carries the hedge: under the stochastic reading of the routing race, several members are trained on overlapping evidence and the population is passed on as a population. Best-of-n collapses at the boundary; a gaggle on a channel does not, because the consumer of the gaggle is the harness, and the harness is a process runtime. This is the "honest version" of [1, §5.5], built once as substrate.
3. Compile-out moves mechanisms across the line. A member that has been compiled into a clause is no longer a confluent object; it is a race the reproduction note can cross over, and the science it encodes is inherited as science.

The honest summary. The design is a saddle point with a ramp off it. The ramp is compile-out, and the design's success is measured by how much of the gaggle, over time, it manages to replace with clauses. A gaggle that never compiles anything out is the loop architecture with better data; a gaggle that compiles everything out has become the mortal scientist of [11], and the transformers were the scaffolding it climbed.

8 The harness, as a term

The sketch below is in the concrete syntax of f1r3node's rholang with the graded guard and `do` as proposed in [5, 2]; it is schematic in the ground-layer calls.

```
new specimens, candidates, traces, gaggle, revise, run, budget in {

  // Propose: a graded race over experiments
  contract propose() = {
    for( @S <<- specimens & @C <<- candidates
         where [ C in F0 and positive(S, C) ] * gain(S, C) / (1 + cost(S, C))
```

```
      ) { run!(S, C) }
  } |

  // Run: one deployment under the do, record the experimental trace
  for( @S, @C <= run ) {
    do( C ) in { *S } |                  // the specimen, intervened
    for( @h, @u, @starved <- trace_of(S, C) ) {
      traces!((S, C, h, u, starved)) | propose!()
    }
  } |

  // Route: attention over members; starvation spawns
  for( @t <<- traces & @(j, sig, T) <<- gaggle
       where [ sig(t) ] * affinity(T, t)
     ) { revise!(j, t) } |
  for( @t <- traces where [ not(any_sig(t)) ] ) {
    new j in { gaggle!((*j, sig_from(t), fresh_member(t))) | revise!(*j, t) }
  } |

  // Revise: natural-gradient step with the per-site count as metric
  for( @j, @t <= revise ) {
    for( @(j, sig, T) <- gaggle & @n <- count(j, site(t)) ) {
      gaggle!((j, sig, natgrad_step(T, t, n))) | count(j, site(t))!(n + 1)
    }
  } |

  // Shrink: merge, prune, compile out, starve  (each a standing race)
  for( @(j, s, Tj) <<- gaggle & @(k, s, Tk) <<- gaggle
       where [ j != k ] * bisim_evidence(Tj, Tk) ) { merge!(j, k) } |
  for( @(j, s, T) <<- gaggle & @psi <<- clause_candidates(j)
       where interventional_fit(T, psi) ) { compile_out!(j, psi) } |
  for( @(j, s, T) <<- gaggle where [ earnings(j) < rent(j) ] ) { starve!(j) }
}
```

Everything the harness decides is a receipt with a clause, so everything it decides can be intervened upon by a harness one level up, which is the recursion that self-improvement needs. The sketch does not show the budget flowing through each receipt; in the cost-accounted calculus every COMM is priced and the seal tree of [16] is where the flow is read off.

9 Execution

On f1r3node. The transformer proposal's Levels 0–2 are what the gaggle needs: a ground tensor type, graded guards in the pure evaluator, and the log-semiring resolver. The experimental trace of Definition 7 is what the order-then-execute design already records — the executing validator resolves the races and the winners are logged; the losers are recomputable from the pre-state. The do lowers at elaboration under $\mathcal{T}_2$. Nothing in §5 needs a construct the two companion proposals have not already asked for, and the harness's own races are ordinary graded receipts.

On Saph1r3. The Mat machine fires a maximal independent set of the conflict graph per step, and the graded machine instead resolves each conflict component by its algebra. The gaggle is a favourable workload: each member's attention contention sets are star-shaped (one query, many keys) and small; the routing race's contention set has one candidate per member; and the compile-out race is embarrassingly parallel across members. The three items the

earlier proposal listed as missing from the route — ground atoms in the Mat state, peek as a primitive, and weights on the conflict graph — are the same three here, and the fourth, segmented reduction for the path-sum reading, is a standard kernel. The route is nascent and the note does not pretend otherwise; what it adds is that a gaggle of small members is a better first weighted workload than one large one, because its contention sets are known statically and its parallel step count is the depth of the harness's loop rather than the width of the gaggle.

The budget. [4, Rem. 6.5] asks that the denotation's budget β, the risk ledger's ω, the address-space bound, and the inference-token budget be one budget doing several jobs. The gaggle adds the parameter budget to the list: tokens buy parameters, parameters are paid for by the score their traces earn, and the whole is one ledger.

10 Obligations

Obligation 1 (The intervention-indexed family is rung two)**.** Proposition 8 is stated on the staged fragment with $\mathcal{F}_0$ constraints. The general statement should say, for a term off $\mathcal{T}_1$, which constraint set $\mathcal{C}$ makes the family $\{D(C) : C \in \mathcal{C}\}$ determine the term up to probabilistic bisimulation, and should connect the answer to the branching cells of [18] as the interventions note's own Obligation 2 proposes. Everything in §6 rests on this.

Obligation 2 (Merge soundness)**.** Show that replacing two interventionally ϵ-bisimilar members on constraint set $\mathcal{C}$ by one preserves the gaggle's interventional distributions on $\mathcal{C}$ to a stated tolerance, and characterise the tolerance's growth off $\mathcal{C}$.

Obligation 3 (Compile-out search)**.** Give the grammar of $\mathcal{F}_1$ candidates generated from a member's attention structure, and the acceptance test, as a graded race with a stated budget; measure on the vending-machine benchmark and on the two laddered games of [10] how much of a gaggle compiles out and how fast.

Obligation 4 (Evidence conservation under distillation)**.** Principle 15 counts distillation targets at zero. The testimony experiment of [3, §11] should be re-run with a transformer-backed member in place of one clause to confirm that the plasticity-loss pathology, and the remedy, transfer.

Obligation 5 (The parameter bound off the finite fragment)**.** Proposition 17 is a CPT count. For continuous or term-valued domains, where a transformer is needed, the bound should be restated in terms of the member's capacity and the mechanism's description length in the clause language, which is where compile-out's acceptance threshold comes from.

Obligation 6 (Experimental design as a theorem)**.** The gain in §5.2 is the expected local score. The claim that choosing experiments by it is information-optimal under the budget should be stated against the optimal-experimental-design literature and against the experimental-regimes note [14], whose interaction-only regimes provably cannot certify universal negatives at polynomial cost — a gaggle that only ever intervenes without reflecting inherits that limit.

Obligation 7 (The recursion)**.** §8 observes that every decision of the harness is a receipt a harness one level up may intervene on. State the fixed point: a harness whose specimens include its own quotation, with the budget as the unrolling depth, and say what Observation 5 demands of it that the one-level design does not already supply.

11 Conclusion

A transformer cannot learn branching from traces, and a harness in rho with graded where-clauses and `do` can produce traces that carry their branching. Lashing the two together is not a reconciliation of the inhabiting-choice note with current practice; it is current practice done with the right data and an honest account of where the race lives. The race lives in the harness, the harness is a term, and the term is heritable — that much is new. The members of the gaggle remain confluent, and the design's ambition is to make them temporary: to learn each mechanism with a small transformer whose context is a parent set and whose job is one conditional, and then to compile it out into a clause the moment the harness understands it well enough to write it down.

The parameter question answers itself along the way. A gaggle backing a causal model is as large as the model's mechanisms and no larger; the harness's experiments are what establish the factorisation that makes the mechanisms small; the do-calculus tells the harness which members it may keep apart; intervention on a member's own heads tells it which heads to keep; interventional bisimulation tells it which members are one; and compile-out tells it when a member can go. None of these is a compression technique applied from outside. Each is the same observation — that intervention sees what observation cannot — applied to the gaggle itself.

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References

[1] L. G. Meredith. Inhabiting choice: confluent witnesses, the phenomena they cannot carry, and one observation made three times. F1R3FLY.io research note, November 2026 (`proof-dynamics-and-reasoning/inhabiting-choice`).

[2] L. G. Meredith. Interventions at the cut: Pearl's `do` as a scoped guard in the rho calculus with graded where-clauses. F1R3FLY.io working note, draft 1, November 2026 (`fuzzyware/interventions-at-the-cut`).

[3] L. G. Meredith. Graded where-clauses, II: the non-negative reals, and where the mortal scientist gets gradient descent. F1R3FLY.io, 2026 (`fuzzyware/graded-where-pln`).

[4] L. G. Meredith. Graded where-clauses, version 4. F1R3FLY.io, 2026 (`fuzzyware/graded-where-v4`).

[5] L. G. Meredith (with Claude). A transformer in f1r3lang: an implementable model. F1R3FLY.io proposal, September 2026.

[6] L. G. Meredith. Name-free combinators for concurrency, draft 3. F1R3FLY.io, 2026 (`rho-combinators/rho-combinators-draft3`).

[7] L. G. Meredith. The rho combinators as a matrix: a data-parallel representation of the reduction graph, v1.2. F1R3FLY.io, 2026 (`rho-combinators/rhocomb-matrix`).

[8] L. G. Meredith. F1R3Comb-Mat v0.5: a bulk-synchronous matrix machine. F1R3FLY.io, 2026 (`rho-combinators/f1r3comb-mat-spec-v05`).

[9] F1R3FLY.io. Saph1r3: compiler from rho to GPU via rho combinators to matrix representation. https://github.com/F1R3FLY-io/Saph1r3, revision 205cdfb, 2026.

[10] L. G. Meredith. Composing learners: a learner is a namespace. F1R3FLY.io, 2026 (`rho-life/compose-note`).

[11] L. G. Meredith. The mortal scientist: learning as cost-accounted computation in a reflective higher-order calculus. F1R3FLY.io, 2026 (`rho-life/scientist-note`).

[12] L. G. Meredith. Reproduction: reflection-based genetics for mortal scientists. F1R3FLY.io, 2026.

[13] L. G. Meredith. Proof search as concurrent reaction. F1R3FLY.io, 2026 (`proof-dynamics-and-reasoning/proof-search`).

[14] L. G. Meredith. Experimental regimes: laboratories for universal negatives, v3. F1R3FLY.io, 2026 (`pnp-experiments/experimental-regimes-v3`).

[15] L. G. Meredith. Graph-structured lambda theories: a reference for the working developer. F1R3FLY.io, 2026.

[16] L. G. Meredith. Cost accounting as a monad. F1R3FLY.io, 2026 (`cost-accounting-as-monad`).

[17] K. G. Larsen and A. Skou. Bisimulation through probabilistic testing. *Information and Computation* 94(1):1–28, 1991.

[18] S. Abbes and A. Benveniste. True-concurrency probabilistic models: branching cells and distributed probabilities for event structures. *Information and Computation* 204(2):231–274, 2006.